

Problem Set 2

Due: Saturday, February 10

We say that a class $\mathcal{F}$ of $\{0, 1\}$ -valued functions defined on $\mathcal{X}$ shatters $\{x_1, \dots, x_n\} \subset \mathcal{X}$ if the cardinality of the set $\mathcal{F}(x_1^n) := \{(f(x_1), \dots, f(x_n)) : f \in \mathcal{F}\}$ is the maximum possible, i.e. $|\mathcal{F}(x_1^n)| = 2^n$. The VC dimension of $\mathcal{F}$ is

$$d_{VC}(\mathcal{F}) = \sup \{n : \mathcal{F} \text{ shatters some } \{x_1, \dots, x_n\} \subset \mathcal{X}\}$$

i.e., the largest n such that there is an n -subset of $\mathcal{X}$ that is shattered by $\mathcal{F}$.

The VC dimension of a collection of subsets of $\mathcal{X}$ is the VC dimension of the class of indicator functions of those sets in the sense defined above.

Problem 1.1

This problem investigates the empirical performance of the perceptron and related algorithms. Throughout, generate $n = 200$ data points in $\mathbb{R}^2 \times \{-1, 1\}$ such that the classes are linearly separable by $\{x : \theta^T x = 0\}$ with $\theta = (-1, 2)$, with margin $\gamma = 1$, and X otherwise uniform in $[-R, R]^2$ with $R = 10$. Use Monte Carlo simulation over $N = 50$ batches to compute the excess risks. In each case, run the competing algorithms on the same training and test data.

- (a) Implement the perceptron algorithm with output $\hat{f}_n$ and the version we called $\tilde{f}_n$ in Theorem 4 in the slides. Empirically obtain the distribution of their excess risks (i.e., $\mathcal{E}(\hat{f}_n)$ and $\mathcal{E}(\tilde{f}_n)$) and compare the mean and variances of these two distributions.
- (b) Implement the perceptron and the SGD version of the logistic regression (as discussed in the slides) with step size $\alpha = 1$. Compare the two algorithms in terms of the speed of convergence and the distribution of the excess risk.

Problem 1.2

- (a) Show that the minimax lower bound derived in the lecture holds if we replace the class of linear threshold functions ($\mathcal{F}_{\text{t-lin}}$) in the definition of $\mathcal{P}$, with any class $\mathcal{F}$ with $d_{VC}(\mathcal{F}) \geq d$.
- (b) Derive a minimax lower bound on the excess risk for the class $\mathcal{F}$ consisting of the indicator functions of boxes in $\mathbb{R}^2$.

Problem 1.3

Show that the class of all closed convex sets in $\mathbb{R}^d$ has infinite VC dimension.

Hint: Consider the set of n points on the boundary of the unit ball.

Problem 1.4

Consider the class of all spheres in $\mathbb{R}^2$, that is,

$$\mathcal{S}_{\text{sph}}^2 = \{S_{a,b} : (a, b) \in \mathbb{R}^2 \times \mathbb{R}_+\}, \quad \text{where} \quad S_{a,b} = \{x \in \mathbb{R}^2 : \|x - a\| \leq b\}.$$

- (a) Show that $\mathcal{S}_{\text{sph}}^2$ can shatter any subset of three points that are not collinear.
- (b) Show that no subset of four points can be shattered, hence $d_{VC}(\mathcal{S}_{\text{sph}}^2) = 3$.

Problem 1.5

Consider the classification problem setup with $\mathcal{Y} = \{-1, 1\}$. For some cost function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ let us define the ϕ -approximation error over the class of functions $\mathcal{F}$ as

$$\mathcal{E}_{\text{ap}}(\mathcal{F}; \phi) = \inf_{f \in \mathcal{F}} \mathbb{E} \phi(Yf(X)) - \inf_f \mathbb{E} \phi(Yf(X))$$

where the second infimum is over all measurable functions. Note that $f : \mathcal{X} \rightarrow \{\pm 1\}$.

Let $\phi_2(t) = (1 - t)^2$ be the quadratic cost and $\phi_{2,+}(t) = (\max(1 - t, 0))^2$ which is the cost function used in two-norm SVM. Also, let $\phi_*(t) = 1\{t < 0\}$.

- (a) Argue that $\inf_f \mathbb{E} \phi_*(Yf(X))$ is the optimal Bayes risk under the 0-1 loss.
- (b) Give an upper bound on $\mathcal{E}_{\text{ap}}(\mathcal{F}; \phi_*)$ in terms of $\mathcal{E}_{\text{ap}}(\mathcal{F}; \phi_2)$ as well as $\mathcal{E}_{\text{ap}}(\mathcal{F}; \phi_{2,+})$.
- (c) How do the ϕ -approximation errors compare?

Problem 1.6

In this problem, we consider bounded regression which can be thought of as a learning problem with $\mathcal{X} \subset \mathbb{R}^d$ and $\mathcal{Y} \subset [-b, b]$ for some $b > 0$. Consider the empirical risk minimization with quadratic loss $\ell(a, y) = (a - y)^2$ over a finite function class $\mathcal{F}$ of functions from $\mathcal{X}$ to $\mathcal{Y}$ (in particular, these are uniformly b -bounded functions).

Show that ERM $\hat{f}_n$ achieves the fast rate for the excess risk, i.e. w.p. $\geq 1 - \delta$, we have

$$\mathcal{E}(\hat{f}_n) \leq C_b \frac{\log(|\mathcal{F}|/\delta)}{n}$$

where $C_b > 0$ is a constant that depends on b .

(Hint: use Bernstein's inequality and bound the variance in terms of the mean similar to what we did in the lecture.)